Bagging

Bootstrap aggregating of regression trees; variance, bias, and the contrast with boosting

Data: eight points

Summary.
1. A fully grown decision tree fits its training data very closely, so on average it is not far from the truth (*low bias*), but a different training sample gives a very different tree (*high variance*).
2. **Variance** is how much the prediction at a fixed point moves when the training sample changes.
3. If we average many models whose mistakes are not all the same, part of the variance cancels, and the average of the predictions stays where it was.
4. We cannot collect many fresh training sets, so we make many resampled ones from the one we have: the **bootstrap**.
5. **Bagging** (*bootstrap aggregating*) trains one tree on each resample and averages them (or takes a vote). It lowers the variance and leaves the bias about where it was; boosting does the opposite job, lowering the bias.

Colour key: every symbol keeps its colour in every formula

μ	**the true value**	$\mu(x)$: what the prediction should be at x (we know it here; in real life we never do)
f	**the prediction**	$\hat{f}(x)$: what one model predicts at x
ε	**the noise**	the random scatter of a data point around its true value
Bias	**the bias**	how far the *average* prediction is from the true value
σ	**the spread**	σ^2 is the variance: how much a prediction moves between training sets
B	**the number of trees**	how many trees we average
ρ	**the correlation**	how much two trees' predictions move together (0 = independent, 1 = identical)

1. The problem: a tree that memorises

Regression means predicting a number. Here is a small, *made-up, illustrative* data set: how long one training epoch takes (minutes) for a neural network of a given size.

point i	model size x (millions)	time y (minutes)	true value $\mu(x) = x + 1$	noise $y - \mu(x)$
1	1	2	2	0
2	2	3	3	0
3	3	3	4	−1
4	4	5	5	0
5	5	6	6	0
6	6	8	7	+1
7	7	8	8	0
8	8	9	9	0

In this lesson *we* know how the data were made: the true time is $\mu(x) = x + 1$, and each measurement is the true value plus a little random noise ε. (In real life the truth is never known. We know it here

so that we can measure how good each method is.) Points 3 and 6 happen to be pushed down and up by 1 minute.

The model. A **fully grown tree** keeps asking yes/no questions about x until every leaf holds a single point. In one dimension this has a simple consequence: its prediction at a new x_0 is the y of the **nearest training point** (the questions are placed halfway between neighbouring points). On the training points it is perfect: the error is 0. The training error is zero because the tree has also fitted the noise; this is the problem.

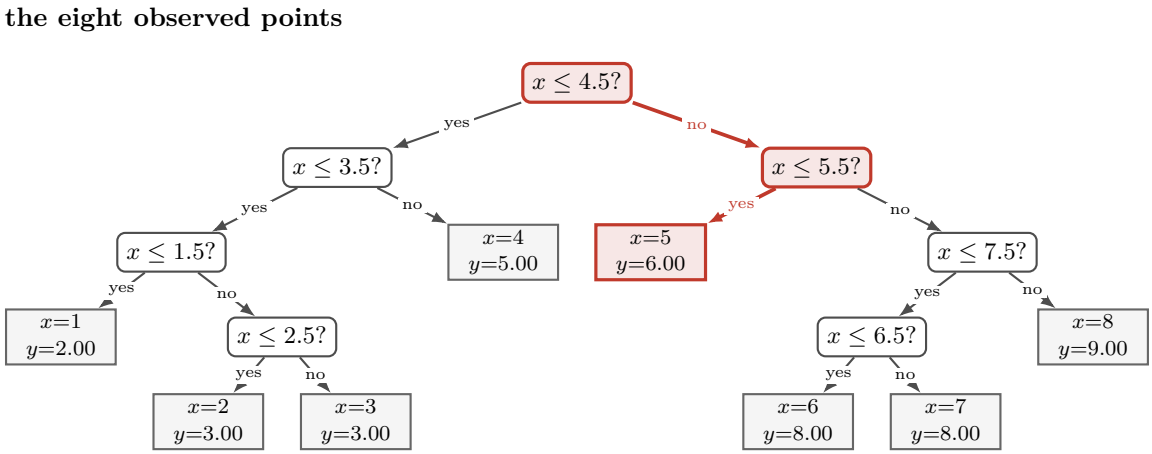


Figure 1. A fully grown tree for the eight observed points. Each rounded box is a yes/no question about x (left branch: yes); each leaf holds one training point and its y, and that y is the prediction for every x that ends there. Red: the path of the new point $x_0 = 4.6$. It answers the questions from the top ($4.6 \le 4.5$? no; $4.6 \le 5.5$? yes) and ends in the leaf of $x = 5$, so the prediction is 6.00.

Example. At $x_0 = 4.6$ the nearest training point is $x = 5$, so the tree predicts $y_5 = 6.0$, while the true value is $\mu(4.6) = 5.6$. At $x_0 = 6.4$ the nearest point is $x = 6$, so it predicts $y_6 = 8$; the truth is 7.4: it has copied the $+1$ noise of point 6.

Your turn 1: read the tree

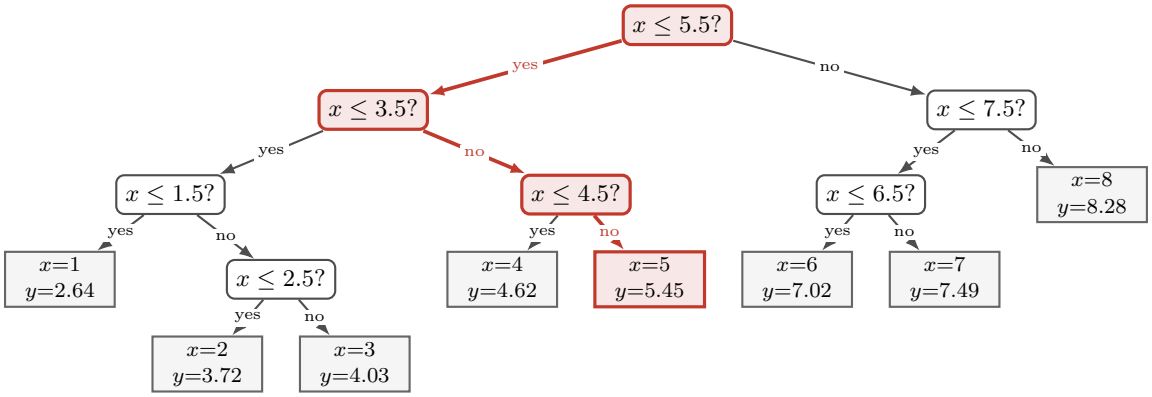
What does the fully grown tree of the table predict at $x_0 = 2.7$ and at $x_0 = 3.4$? What are the true values $\mu(x_0)$? Which of the two predictions is further from the truth, and why?

Hint: nearest training point: 2.7 *is nearest to* 3, 3.4 *is nearest to* 3 *too. The point at* $x = 3$ *has noise* -1.

2. What variance means

What "prediction variance" means. Fix one point x_0. The tree's prediction at x_0 is one number: follow the questions from the top of the tree to a leaf and read off the leaf's value (Figure 1). Now suppose the training data had come out a little differently. The tree would be a different tree, x_0 would end in a different leaf, and the prediction would be a different number (Figure 2). **Prediction variance is how much this one number varies when only the training set changes.** It is *not* the scatter of the data y around the truth, and it is not how the prediction changes from one x to the next.

training set 1



training set 3

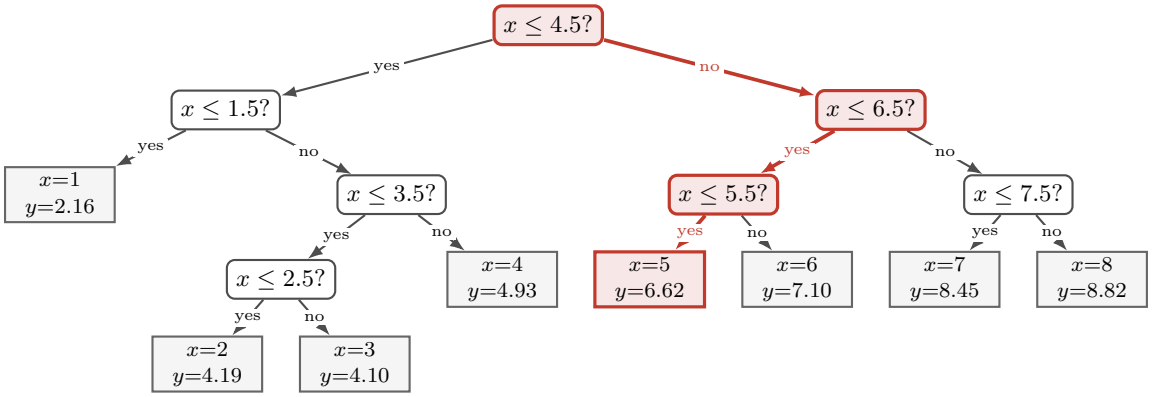


Figure 2. Two other training sets from the same source (same x values, new noise): the ones with the lowest and the highest of the six predictions in the table below. The trees differ in their questions and in their leaf values, and the same point $x_0 = 4.6$ (red path) gets a different prediction: 5.45 against 6.62. **That change of the prediction, when only the training set changes, is the prediction variance.**

The thought experiment. Imagine we could run the experiment again and again: each time, a new training set of eight points (same x's, fresh noise), fit the tree, and look at its prediction at the same $x_0 = 4.6$. Each fit gives a different number. The six below are an actual simulation:

training set	prediction at $x_0 = 4.6$ (y at $x = 5$)	deviation from the average	squared deviation
1	5.45	−0.50	0.253
2	6.00	+0.05	0.002
3	6.62	+0.67	0.444
4	5.46	−0.49	0.243
5	6.33	+0.38	0.142
6	5.86	−0.09	0.009
average	5.953		0.182

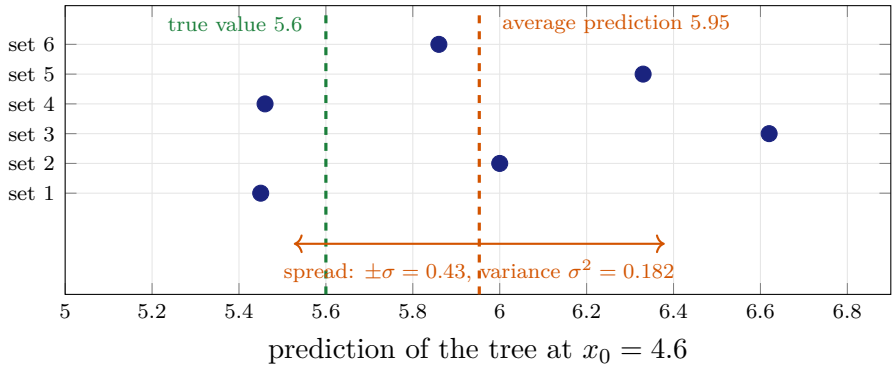


Figure 3. The six predictions of the table, as dots on a number line. The *variance* measures how far the dots are spread around their own average (orange); the *bias* is the distance from that average to the true value (green).

The **average prediction** over the training sets is $\bar{f} = 5.953$. The spread around it, the **variance**, is the average of the squared deviations:

$$\mathrm{Var}(\hat{f}(x_0)) = \mathbb{E}\Big[\big(\hat{f}(x_0) - \mathbb{E}[\hat{f}(x_0)]\big)^2\Big] \approx \frac{\text{sum of squared deviations}}{6} = 0.182.$$

(Six sets are few, so this is only a rough estimate. Over many sets it would settle near $\mathrm{Var}(\varepsilon) = 0.5^2 = 0.25$, because the tree simply copies y_5.) **Variance does not say whether the prediction is right; it says how much it would change if the data changed.**

The other half of the story: bias. If the average prediction $\mathbb{E}[\hat{f}(x_0)]$ is far from the true value $\mu(x_0)$, the model is systematically wrong. This systematic error is the **bias**:

$$\mathrm{Bias}(\hat{f}(x_0)) = \mathbb{E}[\hat{f}(x_0)] - \mu(x_0).$$

Here $\mathbb{E}[\hat{f}(4.6)] = \mathbb{E}[y_5] = \mu(5) = 6$, against $\mu(4.6) = 5.6$: a small bias of 0.4 (the tree has to pick a neighbour's value). A fully grown tree has very low bias: it can follow any shape.

How both add up. Take a fresh measurement $y_0 = \mu(x_0) + \varepsilon_0$ at the same point and ask for the expected squared error of the prediction:

$$\boxed{\mathbb{E}[(y_0 - \hat{f}(x_0))^2] = \mathrm{Bias}^2 + \mathrm{Var}(\hat{f}(x_0)) + \mathrm{Var}(\varepsilon)}$$

Three kinds of error: the **bias** (the model is systematically off), the **variance** (the model moves with the sample) and the **noise** (the data scatter, which nobody can predict). We can only reduce the first two. Box D1 derives this.

Your turn 2: compute a variance

A model gives these predictions at the same point for five training sets: 2.9, 5.1, 4.2, 3.4, 5.4. Find the average prediction, the five deviations from it, and the variance (the average squared deviation).

Hint: average $= 21.0/5$*; then square each deviation and average the five squares.*

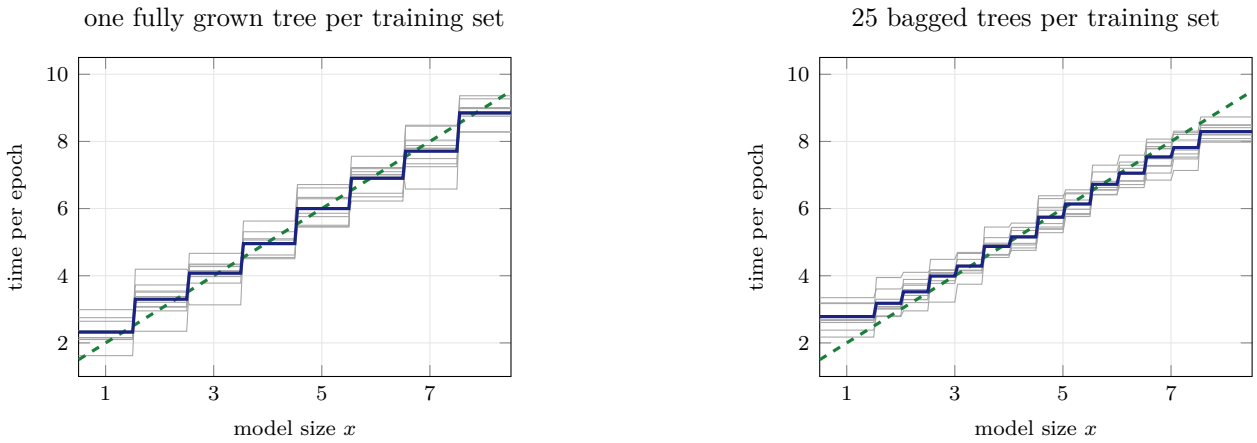


Figure 4. Ten different training sets of eight points from the same source (thin grey: the fit from one training set; dashed green: the true value; thick blue: the average of the ten fits). Left: a fully grown tree jumps about from set to set, but its average follows the truth: *low bias, high variance.* Right: bagging the trees with 25 resamples per training set keeps the grey curves much closer together: *much lower variance.*

3. Why averaging reduces variance

Averaging independent predictions. Suppose we have B trees, each with the same prediction variance σ^2, and suppose their predictions are **independent**. Their average is $\bar{f} = \frac{1}{B}\sum_{b=1}^{B} \hat{f}_b(x_0)$. A constant factor comes out of a variance squared, and the variance of independent terms adds up:

$$\mathrm{Var}(\bar{f}) = \frac{1}{B^2}\sum_{b=1}^{B} \mathrm{Var}(\hat{f}_b) = \frac{1}{B^2}\, B\, \sigma^2 = \boxed{\frac{\sigma^2}{B}}$$

Five independent trees with $\sigma^2 = 0.25$ give $0.25/5 = 0.05$: one fifth of the variance. And the **bias** does not move: the average of the predictions has the average of their expectations, $\mathbb{E}[\bar{f}] = \frac{1}{B}\sum_b \mathbb{E}[\hat{f}_b]$, which equals the single tree's $\mathbb{E}[\hat{f}]$ when all trees have the same expected prediction. **Averaging changes the variance and leaves the bias alone.**

The catch: the trees are not independent. Trees trained on resamples of the same data look alike. Let every pair of trees have correlation ρ (between 0 and 1). Then each of the B variances contributes σ^2, and each of the $B(B-1)$ ordered pairs contributes the covariance $\rho\sigma^2$:

$$\mathrm{Var}(\bar{f}) = \frac{1}{B^2}\left[B\sigma^2 + B(B-1)\rho\sigma^2\right] = \boxed{\rho\,\sigma^2 + \frac{(1-\rho)\,\sigma^2}{B}}$$

Check the two ends: $\rho = 0$ gives σ^2/B (independent trees); $\rho = 1$ gives σ^2 (identical trees: averaging copies does nothing). With $\sigma^2 = 1$:

number of trees	$\rho = 0$	$\rho = 0.3$	$\rho = 0.6$	$\rho = 1$
$B = 1$	1.000	1.000	1.000	1.000
$B = 4$	0.250	0.475	0.700	1.000
$B = 25$	0.040	0.328	0.616	1.000
$B = \infty$	0.000	0.300	0.600	1.000

Read the table by rows. Averaging helps most when the trees disagree in unrelated ways (small ρ). However many trees we add, the variance cannot fall below $\rho\sigma^2$: **the floor is set by how alike the trees are**. This is the reason for the bootstrap and, later, for random forests.

Your turn 3: averages of trees

Each tree has prediction variance $\sigma^2 = 0.25$. Find the variance of the average of $B = 4$ trees (a) if the trees are independent, (b) if every pair has correlation $\rho = 0.5$.

Hint: (a) σ^2/B. *(b)* $\rho\sigma^2 + (1-\rho)\sigma^2/B$.

4. The bootstrap: many training sets from one

Averaging needs many different trees, but we have only *one* training set. The **bootstrap** makes new training sets from it: draw N points **with replacement** from the N we have. A point can appear twice or three times, and some points do not appear at all. Each draw is a new, slightly different, training set of the same size.

Example: five bootstrap samples of the eight points. (Each sample was drawn at random: for each of the eight places, pick one of the eight points with equal chance.)

sample	points drawn (sorted)	left out	different points
1	1, 1, 2, 2, 3, 6, 6, 7	4, 5, 8	5
2	1, 2, 2, 2, 4, 4, 7, 7	3, 5, 6, 8	4
3	1, 1, 1, 2, 4, 4, 7, 7	3, 5, 6, 8	4
4	1, 2, 3, 3, 3, 5, 5, 7	4, 6, 8	5
5	1, 2, 2, 2, 4, 4, 6, 8	3, 5, 7	5

Look at sample 1: some points are drawn more than once and others are missing. The points that are left out of a sample are its **out-of-bag** points. How many are left out on average? The chance that one given point is *not* picked in a single draw is $1 - \frac{1}{N}$; in N independent draws it is never picked with probability $\left(1 - \frac{1}{N}\right)^N$:

N	chance a point is left out	share of the points that appear
8	0.344	0.656
20	0.358	0.642
100	0.366	0.634
1000	0.368	0.632
∞	$e^{-1} = 0.368$	0.632

So a bootstrap sample contains about 63% of the different points, and about 37% are left out (Box D4 shows the limit e^{-1}). Each tree therefore sees a different part of the data, which is what makes the trees different.

Your turn 4: leaving a point out

For $N = 8$ compute the chance that point 1 is not drawn in any of the 8 draws, $\left(1 - \frac{1}{8}\right)^8$. Use $(7/8)^2 = 0.7656$, then square twice.

Hint: $(7/8)^8 = \left((7/8)^2\right)^4$.

5. Bagging, step by step

Bagging for regression. For $b = 1, \ldots, B$: draw a bootstrap sample, grow a tree on it (fully grown, no pruning). To predict at x, **average** the B predictions: $\hat{f}_{\text{bag}}(x) = \frac{1}{B} \sum_{b=1}^{B} \hat{f}_b(x)$.

Worked example at $x_0 = 4.6$, with $B = 5$. Each tree is fully grown on its own sample, so (as before) it predicts the y of the nearest point *that is in its sample*. The true value is $\mu(4.6) = 5.6$.

tree	different x in its sample	nearest to 4.6	prediction $\hat{f}_b$
1	1, 2, 3, 6, 7	6	8.0
2	1, 2, 4, 7	4	5.0
3	1, 2, 4, 7	4	5.0
4	1, 2, 3, 5, 7	5	6.0
5	1, 2, 4, 6, 8	4	5.0
		average of the five predictions	**5.80**

1 **Step 1.** Draw the five samples (above). 2 **Step 2.** Grow a tree on each: the predictions are 8.0, 5.0, 5.0, 6.0, 5.0. 3 **Step 3.** Average them: $\hat{f}_{\text{bag}}(4.6) = 5.80$.

Compare with the single tree grown on the original eight points, which predicts 6.0 (the truth is 5.6). Different samples give different answers (the variance); the average is steadier. A tree trained on a sample is individually *worse* than a tree trained on all points (it has seen fewer different points), but the errors partly cancel in the average.

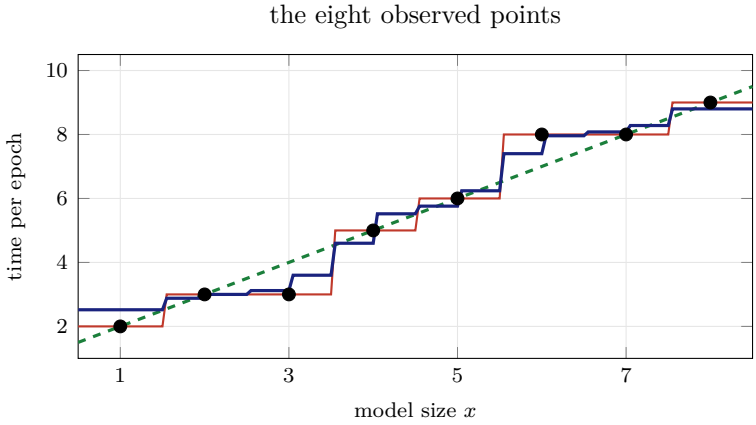


Figure 5. Red: the fully grown tree fitted to the eight observed points (it copies the noise at $x = 3$ and $x = 6$). Blue: the average of 25 bootstrap trees. Dashed green: the true value $\mu(x) = x + 1$. The blue curve is smoother and stays nearer the green line.

Your turn 5: add a sixth tree

A sixth bootstrap sample contains the different points $x = 3,\ 4,\ 5,\ 6,\ 7,\ 8$ (some drawn more than once). Which of them is nearest to 4.6, so what does the sixth tree predict? What is the average of all six predictions?

Hint: the six predictions are the five of the table plus the new one.

Why not train B trees on the same data? They would be identical ($\rho = 1$) and the average would equal one tree. Resampling is what makes the trees differ.

How many trees? More trees never hurt: the variance only falls, towards its floor. (Table 4 in the next section shows this.) **Free validation.** Each tree has not seen about 37% of the points; predicting every point with only the trees that did not see it gives an estimate of the test error, the **out-of-bag error**, with no separate test set.

6. Bias and variance: what bagging fixes, what boosting fixes

We can now measure the three kinds of error. We simulated 4000 training sets of eight points (true value $x+1$, noise standard deviation 0.5, so $\mathrm{Var}(\varepsilon) = 0.25$). For each method we computed, averaged over the eight values of x, the squared bias, the variance, and their sum plus the noise (the expected test error):

method	bias^2	variance	noise	expected error
average of the y's only (no tree)	5.250	0.033	0.250	**5.533**
boosting, 1 round ($\nu = 0.5$)	2.104	0.222	0.250	**2.575**
boosting, 3 rounds	0.426	0.268	0.250	**0.944**
boosting, 8 rounds	0.011	0.261	0.250	**0.522**
one fully grown tree	0.000	0.252	0.250	**0.502**
bagging, 25 fully grown trees	0.081	0.146	0.250	**0.477**

But each tree in boosting is itself high bias. How can the sum have low bias? Bias belongs to the *whole model*, not to its pieces, and boosting does not ask a stump to fit y. Each new stump is fitted to the **residual**: whatever the current model still gets wrong. The model is the *sum* of all stumps so far. Two things follow:

- One stump has two levels, so it can never be a straight line: alone it is badly biased. A sum of t stumps can have up to $t+1$ levels, so it can follow the truth more and more closely: the family of functions grows with every round.
- At each round the stump has an easier job than fitting y: it only has to remove part of the remaining error. It need not be good, only better than nothing on the residual (this is what "weak" means). The error that remains after the round is smaller, and the next stump works on that.

Example without noise, so that only the bias is seen: the truth is $\mu(x) = x + 1$ at $x = 1, \ldots, 8$, the start is the average (5.5), and each round adds one stump fitted to the residual, with full step size. The last column is the mean squared distance between the model and the truth, which is the squared bias.

round	stump added: question, left value, right value	$F(1)$	$F(2)$	$F(3)$	$F(4)$	$F(5)$	$F(6)$	$F(7)$	$F(8)$	bias^2
0	start: the average	5.50	5.50	5.50	5.50	5.50	5.50	5.50	5.50	5.25
1	$x \le 4.5$: -2.00, $+2.00$	3.50	3.50	3.50	3.50	7.50	7.50	7.50	7.50	1.25
2	$x \le 2.5$: -1.00, $+0.33$	2.50	2.50	3.83	3.83	7.83	7.83	7.83	7.83	0.92
3	$x \le 7.5$: -0.17, $+1.17$	2.33	2.33	3.67	3.67	7.67	7.67	7.67	9.00	0.72
4	$x \le 4.5$: $+0.50$, -0.50	2.83	2.83	4.17	4.17	7.17	7.17	7.17	8.50	0.47

Round 1: the residuals are $-3.5, \ldots, +3.5$; the question $x \le 4.5$ puts the four negative residuals on one side (average -2) and the four positive ones on the other (average $+2$), so the model becomes 3.5 on the left and 7.5 on the right, and the squared bias falls from 5.25 to 1.25. Round 2 looks at

what is left (the left half is now too high at $x = 1, 2$ and too low at $x = 3, 4$) and fixes part of it. Each round lowers the squared bias; the stumps are weak, but the sum is not.

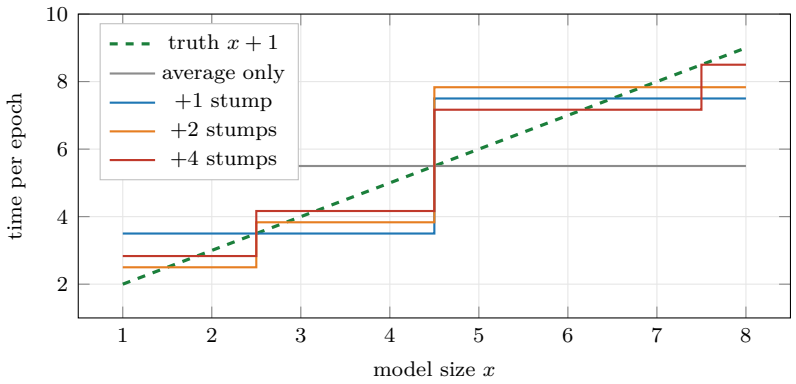


Figure 6. Boosting on noise-free data: the model after 0, 1, 2 and 4 stumps against the truth. Every stump added is a two-step function, yet the sum has more steps each time and follows the line more closely.

Read it from top to bottom.

- **Too simple: high bias.** Predicting just the average of the y's ignores x: the squared bias is 5.25 and the variance is tiny. This is where **boosting** starts.
- **Boosting attacks the bias.** Each round adds a small tree fitted to what is still wrong. After 1, 3 and 8 rounds the squared bias falls from 5.25 to 2.10, 0.43 and 0.011. The price is variance: it rises from 0.033 to 0.261 at 8 rounds, about the level of a fully grown tree, because by then the boosted model can follow the data as closely as that tree. Bias is bought with flexibility, which is why boosting is stopped early. (How a sum of weak trees can have low bias is explained next.)
- **A fully grown tree has the opposite problem: no bias, high variance** (0.252 against a noise of 0.25: the tree copies the noise of every point).
- **Bagging attacks the variance.** Averaging 25 such trees cuts the variance from 0.252 to 0.146, while the squared bias stays tiny (0.081; the bootstrap trees see slightly fewer points, so they are a little more biased).

Note on the totals. With only eight points the net gain of bagging over one tree is small (the last column), because it trades a little bias for much less variance; with more data and noisier trees the gain is larger. The point of the table is *which column* each method moves.

One sentence to remember: *bagging averages away the variance of deep trees; boosting adds up small trees to remove the bias of shallow ones.*

How the correlation ρ between two trees is measured. Take two trees, a and b, each grown on its own bootstrap sample of the same training set, and look at their predictions $u = \hat{f}_a(x_0)$ and $v = \hat{f}_b(x_0)$ at one point. Repeat the whole experiment K times (a new training set, then two bootstrap trees on it). This gives K pairs (u_k, v_k). The correlation is the covariance divided by the spread of each prediction, and it is estimated from the pairs by

$$\rho = \frac{\mathrm{Cov}(u, v)}{\sqrt{\mathrm{Var}(u)\ \mathrm{Var}(v)}} \approx r = \frac{\frac{1}{K}\sum_k (u_k - \bar{u})(v_k - \bar{v})}{\sqrt{\frac{1}{K}\sum_k (u_k - \bar{u})^2 \cdot \frac{1}{K}\sum_k (v_k - \bar{v})^2}}.$$

Reading it: when tree a predicts above its own average, does tree b usually do so too? If yes, the products $(u_k - \bar{u})(v_k - \bar{v})$ are mostly positive and $\rho > 0$. If the two are unrelated, positive and negative products cancel and $\rho \approx 0$. If the trees were identical, $v = u$ and $\rho = 1$. Six pairs at $x_0 = 4.6$:

set k	u_k	v_k	$u_k - \bar{u}$	$v_k - \bar{v}$	product	$(u_k - \bar{u})^2$	$(v_k - \bar{v})^2$
1	5.95	5.95	+0.19	+0.11	+0.02	0.037	0.011
2	6.16	4.91	+0.40	−0.93	−0.38	0.163	0.874
3	5.31	5.05	−0.45	−0.79	+0.36	0.200	0.632
4	6.62	6.62	+0.86	+0.78	+0.67	0.745	0.601
5	5.51	6.53	−0.25	+0.69	−0.17	0.061	0.469
6	4.99	6.01	−0.77	+0.17	−0.13	0.588	0.027
mean / sum/6	5.757	5.845			0.062	0.299	0.436

From the last row, $r = \dfrac{0.062}{\sqrt{0.299 \times 0.436}} = 0.17$. Six pairs are far too few to pin ρ down (a different draw would give a very different r, and this draw was picked to be close to the long-run value so that the table is representative). With $K = 6000$ pairs, averaged over the eight values of x, the same calculation gives $\rho = 0.18$. (This needs a way to generate new training sets, as in a simulation. With real data there is only one training set, so ρ is not measured directly; its effect shows in the way the out-of-bag error stops improving as B grows.)

Your turn 6: a correlation

Five training sets give these predictions of two trees at the same point: $u = 4, 6, 8, 5, 7$ and $v = 5, 5, 7, 6, 7$. Compute $\bar{u}$, $\bar{v}$, the deviations, $\frac{1}{K}\sum(u - \bar{u})(v - \bar{v})$, the two variances, and the correlation r.

Hint: $\bar{u} = \bar{v} = 6$. *Then* $r = \dfrac{covariance}{\sqrt{var(u)\, var(v)}}$.

Does the formula of Section 3 describe bagging? Take the variance of a single bootstrap tree, $\sigma^2 = 0.683$ (the bagged variance with $B = 1$; it is larger than the variance of a tree on the original data, 0.252, because a bootstrap sample is itself random). The correlation can also be found without forming pairs: solving the formula of Section 3 for ρ gives $\rho = \dfrac{\mathrm{Var}_B - \sigma^2/B}{\sigma^2 - \sigma^2/B}$, and with $B = 100$ this is $\rho \approx 0.18$, in agreement with the direct value above. These two numbers fix the formula $\rho\sigma^2 + (1 - \rho)\sigma^2/B$ completely; the rows $B = 1$ and $B = 100$ were used to fit it, so the rows $B = 5, 10, 25$ are genuine predictions:

trees B	simulated variance	formula $\rho\sigma^2 + (1 - \rho)\sigma^2/B$
1	0.683	0.683
5	0.237	0.237
10	0.181	0.181
25	0.146	0.148
100	0.131	0.131

Table 4. Variance of the bagged prediction against the number of trees. The variance falls quickly and then flattens at the floor $\rho\sigma^2$: the trees are partly alike (they are trained on overlapping data), so more trees cannot remove the part of the variance that all of them share.

Bagging and boosting side by side.

	Bagging	**Boosting**
Trees	deep (each: low bias, high variance); the average keeps the bias, lowers the variance	shallow (each: high bias, low variance); the sum has low bias because each tree fits what is left
Built	independently, each on a bootstrap sample: could be done in parallel	one after another, each on what is still wrong
Combined by	average (regression) or vote (classification)	sum of small corrections
Fixes	the **variance**	the **bias**
More trees	never hurts; variance falls to a floor	too many rounds can start to fit the noise

These are tendencies, not laws. They explain the usual choices: deep trees for bagging, stumps or small trees for boosting.

7. Classification: bagging by voting

For classification each tree predicts a *class*, and the bag predicts the class with the most votes (or averages the trees' probabilities and picks the larger). Why does a vote help? Take a deliberately simple case: every tree is correct with probability 0.6, and the trees are independent. The chance that the **majority** is correct is:

number of voters	chance the majority is correct
1	0.600
5	0.683
25	0.846
101	0.979

Each voter is only slightly better than a coin toss, yet 101 of them are right about 0.979 of the time. This is the variance effect again: one voter's mistakes are random, and the random part cancels in a large majority. (For five voters: the majority is right if at least 3 are right, $\sum_{k=3}^{5} \binom{5}{k} 0.6^k 0.4^{5-k} = 0.683$.)

Two warnings. (1) If the voters all make the *same* mistakes ($\rho = 1$), the majority is no better than one voter: independence is what the table assumed. (2) A vote cannot fix a *bias*: if every voter is wrong on a point for the same systematic reason, so is the majority.

8. One more step: random forests

The floor $\rho\sigma^2$ says: the way to lower the variance further is to make the trees *less alike*. A **random forest** does exactly this on top of bagging: at every split, the tree may only look at a random subset of the features (for example, $\sqrt{d}$ out of d features) instead of all of them. The trees become more different (ρ falls) at the price of each tree being a bit weaker (its variance and bias rise a little). In practice the lower correlation wins.

9. Derive it yourself

The key formulas can be derived in a few lines. Work through the boxes in order; the full working is in the appendix.

Box D1: bias, variance and noise

Let $y_0 = \mu + \varepsilon_0$ with $\mathbb{E}[\varepsilon_0] = 0$, $\mathrm{Var}(\varepsilon_0) = \sigma_\varepsilon^2$, and let $\hat{f}$ be a prediction independent of ε_0 with mean $\bar{f} = \mathbb{E}[\hat{f}]$. Write $y_0 - \hat{f} = (\mu - \bar{f}) + (\bar{f} - \hat{f}) + \varepsilon_0$, square it, take the expectation, and show that it equals $\mathrm{Bias}^2 + \mathrm{Var}(\hat{f}) + \sigma_\varepsilon^2$.

Hint: the three cross terms contain $\mathbb{E}[\bar{f} - \hat{f}] = 0$ *or* $\mathbb{E}[\varepsilon_0] = 0$ *(or both).*

Box D2: independent trees

B independent predictions $\hat{f}_b$ each have variance σ^2. Show that their average has variance σ^2/B.

Hint: $\mathrm{Var}(cX) = c^2\,\mathrm{Var}(X)$*, and for independent terms* $\mathrm{Var}(\sum X_b) = \sum \mathrm{Var}(X_b)$.

Box D3: correlated trees

Now every pair of different trees has covariance $\rho\sigma^2$. Show that $\mathrm{Var}(\bar{f}) = \rho\sigma^2 + \dfrac{(1-\rho)\sigma^2}{B}$. Check $\rho = 0$ and $\rho = 1$.

Hint: $\mathrm{Var}(\sum_b X_b) = \sum_b \mathrm{Var}(X_b) + \sum_{b \neq c} \mathrm{Cov}(X_b, X_c)$*: there are* B *variances and* $B(B-1)$ *covariances.*

Box D4: the 37%

Show that $\left(1-\frac{1}{N}\right)^N \to e^{-1} = 0.368$ as N grows. Then say what share of the different points a bootstrap sample contains.

Hint: take the logarithm: $N\ln\left(1-\frac{1}{N}\right) \approx N\left(-\frac{1}{N}\right) = -1$.

Box D5: why averaging leaves the bias alone

Each tree has $\mathbb{E}[\hat{f}_b(x)] = m(x)$, the same for all trees. Show that the average $\bar{f}$ has $\mathbb{E}[\bar{f}] = m(x)$, so its bias equals the bias of a single tree.

Hint: the expectation of a sum is the sum of the expectations.

10. The recipe and common confusions

Bagging in five lines.

1. Choose the number of trees B (for example 100).
2. For each b: draw N points with replacement from the training set.
3. Grow a fully grown tree on that sample.
4. Predict by averaging the B predictions (regression) or by majority vote (classification).
5. Optionally estimate the test error from the out-of-bag points.

"Why trees?" Fully grown trees have low bias and high variance: they are *unstable* (a small change in the data changes the tree), which is exactly what bagging fixes. A stable model such as linear regression moves little between samples, so bagging would barely help it.

"Does bagging overfit if B is large?" No. More trees lower the variance towards its floor; they do not make the bias worse.

"Is the bootstrap sample cheating, since points repeat?" The repeats only change how much each point counts in one tree. The average over many resamples is what we use, and its error is estimated with the out-of-bag points.

"Bagging or boosting?" If your trees are deep and unstable, average them (bagging, random forest). If your trees are shallow and too simple, add them up (boosting). The two methods fix opposite halves of the error.

11. Check yourself

Fill in the blanks without looking, then check against the answers below.

1. The prediction of a model at a fixed point moves between training sets: this spread is its _____.
2. The average prediction minus the true value is the _____.
3. Expected squared error = _____^{2} + _____+ noise variance.
4. The variance of the average of B independent predictions, each of variance σ^2, is _____.
5. If the trees have correlation ρ, the variance cannot fall below _____, however large B is.
6. A bootstrap sample draws N points _____; about _____ % of the different points appear.
7. A fully grown tree has _____ bias and _____ variance.
8. Bagging mainly reduces the _____; boosting mainly reduces the _____.
9. For classification, bagging combines the trees by a _____; the points a tree has not seen give the _____ error.
10. Adding more trees to a bag lowers the variance towards a _____ set by how _____ the trees are.

Answers

1. variance
2. bias
3. bias; variance
4. σ^2/B
5. $\rho\sigma^2$
6. with replacement; 63
7. low; high
8. variance; bias
9. majority vote; out-of-bag
10. floor; alike (correlated)

Appendix: solutions

Your turn 1

At $x_0 = 2.7$ the nearest training point is $x = 3$ (distance 0.3), so the tree predicts $y_3 = 3$; the true value is 3.7: error 0.7. At $x_0 = 3.4$ the nearest point is again $x = 3$ (distance 0.4; the next point, 4, is 0.6 away): prediction 3, truth 4.4: error 1.4. The second is further off: the tree copies the noise of point 3 ($y_3 = 3$ instead of 4), and the same 3 is used for a longer stretch of x values.

Your turn 2

Average $= 21.0/5 = 4.20$. Deviations: -1.30, 0.90, 0.00, -0.80, 1.20. Squares: 1.6900, 0.8100, 0.0000, 0.6400, 1.4400, sum 4.5800. Variance $= 4.5800/5 = 0.9160$.

Your turn 3

(a) Independent: $0.25/4 = 0.0625$. (b) $\rho = 0.5$: $0.5(0.25) + 0.5(0.25)/4 = 0.125 + 0.03125 = 0.15625$. Correlation makes four trees almost as noisy as two independent ones.

Your turn 4

$(7/8)^2 = 0.7656$; squared: 0.5862; squared again: 0.3436. So point 1 is missing from a given sample of eight with probability 0.344, and appears with probability 0.656.

Your turn 5

The sixth sample contains $x = 3,\ 4,\ 5,\ 6,\ 7,\ 8$. The nearest to 4.6 is 5, so the sixth tree predicts 6.0. Average of the six: $(29.0 + 6.0)/6 = 5.83$ (the five-tree average was 5.80).

Your turn 6

$\bar{u} = (4+6+8+5+7)/5 = 6$ and $\bar{v} = (5+5+7+6+7)/5 = 6$. Deviations of u: $-2,\ 0,\ 2,\ -1,\ 1$; of v: $-1,\ -1,\ 1,\ 0,\ 1$. Products: 2, 0, 2, 0, 1, sum 5, covariance $5/5 = 1$. Variances: $\text{var}(u) = (4+0+4+1+1)/5 = 2$ and $\text{var}(v) = (1+1+1+0+1)/5 = 0.8$. Correlation $r = 1/\sqrt{2 \times 0.8} = 1/1.265 = 0.79$: the two trees tend to move together.

Box D1

Square $y_0 - \hat{f} = (\mu - \bar{f}) + (\bar{f} - \hat{f}) + \varepsilon_0$: the squares are $(\mu - \bar{f})^2$, $(\bar{f} - \hat{f})^2$, ε_0^2, and three cross terms $2(\mu - \bar{f})(\bar{f} - \hat{f})$, $2(\mu - \bar{f})\varepsilon_0$, $2(\bar{f} - \hat{f})\varepsilon_0$. Take expectations. $(\mu - \bar{f})$ is a constant and $\mathbb{E}[\bar{f} - \hat{f}] = 0$, so the first cross term vanishes; $\mathbb{E}[\varepsilon_0] = 0$ kills the second; $\hat{f}$ and ε_0 are independent and each has a zero-mean factor, so the third vanishes too. What remains: $\text{Bias}^2 + \text{Var}(\hat{f}) + \sigma_\varepsilon^2$.

Box D2

$\text{Var}(\frac{1}{B}\sum_b \hat{f}_b) = \frac{1}{B^2}\text{Var}(\sum_b \hat{f}_b) = \frac{1}{B^2}\sum_b \text{Var}(\hat{f}_b) = \frac{B\sigma^2}{B^2} = \frac{\sigma^2}{B}$.

Box D3

$\mathrm{Var}(\sum_b \hat{f}_b) = B\sigma^2 + B(B-1)\rho\sigma^2$, so $\mathrm{Var}(\bar{f}) = \dfrac{\sigma^2}{B} + \dfrac{(B-1)\rho\sigma^2}{B} = \dfrac{\sigma^2}{B} + \rho\sigma^2 - \dfrac{\rho\sigma^2}{B} = \rho\sigma^2 + \dfrac{(1-\rho)\sigma^2}{B}$. For $\rho = 0$ this is σ^2/B; for $\rho = 1$ it is σ^2.

Box D4

$\ln(1 - \frac{1}{N})^N = N \ln(1 - \frac{1}{N})$. For large N, $\ln(1-t) \approx -t$ with $t = \frac{1}{N}$, so this is about $N(-\frac{1}{N}) = -1$, and $(1 - \frac{1}{N})^N \to e^{-1} = 0.368$. So about 36.8% of the points are missing from a bootstrap sample and 63.2% appear.

Box D5

$\mathbb{E}[\bar{f}] = \mathbb{E}[\frac{1}{B}\sum_b \hat{f}_b] = \frac{1}{B}\sum_b \mathbb{E}[\hat{f}_b] = \frac{1}{B} \cdot B \cdot m(x) = m(x)$. The bias is $m(x) - \mu(x)$, the same as for one tree.

Check yourself: answers

1. variance
2. bias
3. bias; variance
4. σ^2/B
5. $\rho\sigma^2$
6. with replacement; 63
7. low; high
8. variance; bias
9. majority vote; out-of-bag
10. floor; alike (correlated)